Universe Unfolding

Exact Causal Realization, Observable Information, and Fourier-Sector Memory

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Abstract

We study how prescribed observable laws admit a single deterministic realization driven by a random seed, uniformly over adaptive choices. Exact reconstruction under one admissible policy fixes the optimal seed entropy. We apply this principle to sum-and-moment memory and to sequential observation of Fourier sectors. A constant or vanishing readout can conceal arbitrarily large finite entropy, while access to a reconstructing sequence of readouts recovers its exact simulation cost. We also prove an exact reduction of almost-surely safe-policy simulation to a viable subtree. These elementary results organize a common framework; no historical priority is claimed. The title denotes unfolding an interaction history, not a derivation of cosmology. A companion causal Shannon bound is stated conditionally on its hypotheses and is not reproved here.

1 Scope and the distinction that matters

Three objects must be kept separate: a state or seed, the laws it realizes, and the interface through which it is observed. Equal observable laws need not mean equal hidden states. A zero readout need not mean zero information in a different readout. Conversely, hidden structure cannot be inferred merely because an output is zero.

Our main results are finite probabilistic statements. Signal labels may belong to an infinite-dimensional function space, but every random signal considered below has finite support. Thus its Shannon entropy is discrete entropy, not differential entropy, signal energy, thermodynamic entropy, or description length of an arbitrary real number. Exact arithmetic and computation time are separate resources.

2 A single seed for an adaptive interface

Let T be a finite tree of complete histories. At a nonterminal history h, an experimenter chooses $a \in A(h)$ from a finite nonempty set and observes $y \in Y(h,a)$ from a finite nonempty set, with probability $p(y \mid h,a)$. These probabilities are nonnegative and sum to one. A deterministic policy π selects actions using only the observed history. It does not inspect the hidden seed.

Definition 2.1 (Exact realization)**.** *A finite seed $S \sim \mu$ and a deterministic decoder f realize a class Π of policies if*

$$A_t = \pi(H_{t-1}), \quad Y_t = f(S, H_{t-1}, A_t), \quad H_t = H_{t-1} \,\|\, (A_t, Y_t), \quad H_0 = \varnothing,$$

induce the prescribed transcript law for every $\pi \in \Pi$. The law μ and the decoder f are fixed independently of the selected policy. Write

$$C_\Pi(T) = \inf_{(\mu,f)\ \text{exact for}\ \Pi} \mathrm{H}(S), \qquad C(T) = C_{\text{all deterministic policies}}(T).$$

The shared decoder enforces agreement at shared histories. The quantifiers are $\exists(\mu, f)\ \forall\pi$, not a separate simulator chosen for each policy. A finite realization always exists: pre-sample independently one output for every history–action pair. Along any trajectory, the next table entry is fresh relative to the entries determining the past. This reproduces the conditional laws by induction.

In the language of the companion project [1], let I_π denote the transcript information $-\log_2 P_T^\pi(H_n)$ on positive-mass transcripts. The lower CDF envelope is

$$G_T(u) = \min_\pi \Pr(I_\pi \le u), \qquad L(T) = \int_0^\infty [1 - G_T(u)]\, du.$$

The companion finite causal Shannon theorem states, on its applicable class,

$$L(T) \le C(T) \le L(T) + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2}. \tag{1}$$

Equation (1) is an imported result, not a premise needed for the direct identities proved below. The latter allow zero probabilities. No general infinite-horizon extension is established in this paper.

3 Exact reconstruction and observational compression

For a finite-valued random variable X and deterministic q, the chain rule gives

$$\mathrm{H}(X) = \mathrm{H}(q(X)) + \mathrm{H}(X \mid q(X)). \tag{2}$$

Hence q preserves entropy precisely when X is recoverable from $q(X)$ almost surely. The decoder can be defined arbitrarily on values outside the support. A nonzero output alone does not guarantee recoverability.

Theorem 3.1 (Reconstructing-policy principle)**.** *Suppose a finite-support random variable Z drives a deterministic causal interface exact for all $\pi \in \Pi$. If one policy π_0 produces a transcript X from which Z is almost surely recoverable, then*

$$\boxed{C_\Pi = \mathrm{H}(Z).}$$

The minimum is attained by the seed $S = Z$.

Proof. The given interface with seed Z gives the upper bound. Since X is a function of Z and conversely Z is recoverable from X, their entropies agree. Under π_0, every competing exact simulator produces a transcript X' with the same law as X, deterministically from its seed U. Thus $\mathrm{H}(U) \ge \mathrm{H}(X') = \mathrm{H}(X) = \mathrm{H}(Z)$. □

Proposition 3.2 (Restriction and recoding)**.** *If $\Pi_1 \subseteq \Pi_2$, then $C_{\Pi_1} \le C_{\Pi_2}$. Invertible relabelings of actions and outputs that respect histories preserve the optimal seed cost.*

Proof. An exact simulator for the larger class also serves the smaller one. For an invertible relabeling, translate policies and decoder outputs in each direction. This gives simulators of identical seed entropy in both directions. □

An unused independent seed component changes no observable law, while increasing seed entropy. Therefore the hidden realization is generally not identifiable; the optimum is an operational resource quantity.

4 Memory that survives cancellation

Fix real addresses $x_1, \dots, x_n$ and complex deposits $Z = (Z_1, \dots, Z_n)$ with an arbitrary finite-support joint law. Define

$$S_0 = M_0 = 0, \qquad S_t = S_{t-1} + Z_t, \qquad M_t = M_{t-1} + x_t Z_t.$$

At time t the observer may request S_t or M_t. Observation is passive: it does not change the deposits or their law.

Corollary 4.1 (Observable writer)**.** *The optimal common seed entropy is*

$$C_n = \mathrm{H}(Z_1, \dots, Z_n) = \sum_{t=1}^{n} \mathrm{H}(Z_t \mid Z_1, \dots, Z_{t-1}).$$

Proof. Always requesting the sum reconstructs $Z_t = S_t - S_{t-1}$. Apply Theorem 3.1 and the chain rule. □

The update algebra is the writer used in the supplied ONE manuscript [7]; the probability law and observation interface are additional inputs. This use of the writer is not an endorsement of the manuscript's cosmological claims.

For two deposits $Z_1 = W$, $Z_2 = -W$ and $x_1 \neq x_2$,

$$S_2 = 0, \qquad M_2 = (x_1 - x_2)W.$$

With terminal-only observation, the sum interface has cost zero, whereas the interface offering either sum or moment has cost $\mathrm{H}(W)$. A constant seed suffices for the first; seed W suffices for the second and its injective moment readout gives the matching lower bound. This is not true of the sum interface if the earlier sum $S_1 = W$ is also observable.

5 Fourier sectors as an exact observation interface

Use the Fourier convention

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi}\, dx.$$

The integral is first defined on Schwartz functions; Plancherel's theorem extends it unitarily to $L^2(\mathbb{R})$. Let h_n be the normalized Hermite basis in this convention, so $h_0(x) = 2^{1/4} e^{-\pi x^2}$ and

$$\mathcal{F} h_n = (-i)^n h_n, \qquad \mathcal{F}^4 = I.$$

These are classical facts, not new consequences of the causal seed theorem. Euler's identity alone is not a proof of them.

For $\Lambda = \{1, -i, -1, i\}$ define

$$P_\lambda = \frac{1}{4} \sum_{j=0}^{3} \lambda^{-j} \mathcal{F}^j.$$

Applying this polynomial to h_n gives h_n if $(-i)^n = \lambda$ and zero otherwise, by the finite geometric-sum identity. Completeness of the Hermite basis consequently yields

$$P_\lambda P_\nu = \mathbf{1}_{\lambda=\nu} P_\lambda, \quad P_\lambda^* = P_\lambda, \quad \sum_{\lambda \in \Lambda} P_\lambda = I.$$

In particular, P_1 is a projection onto the Fourier-fixed subspace, not a universal lossless representation. For example $P_1 h_1 = 0$.

Theorem 5.1 (Sequential Fourier-sector reconstruction)**.** *Let Z be an $L^2(\mathbb{R})$-valued random signal with finite support. In a four-round passive interface, each action requests one of the outputs $P_\lambda Z$, and repeated requests are allowed. The optimal seed entropy for all adaptive policies is exactly* $\mathrm{H}(Z)$*. If only P_1 is available, the optimal seed entropy is* $\mathrm{H}(P_1 Z)$*. The difference is*

$$\boxed{\mathrm{H}(Z) - \mathrm{H}(P_1 Z) = \mathrm{H}(Z \mid P_1 Z).}$$

Proof. The fixed policy requesting each of the four sectors recovers $Z = \sum_\lambda P_\lambda Z$. The seed Z supplies every adaptive sequence of requests. Theorem 3.1 proves the first claim. If only P_1 is available, the first output has entropy $\mathrm{H}(P_1 Z)$ and all repeated outputs equal it. Taking this output itself as seed proves the second claim. Equation (2) gives the difference. □

Corollary 5.2 (A fixed readout with arbitrarily large accessible entropy)**.** *Let W be a finite real random variable and $Z = h_0 + W h_1$. Then*

$$P_1 Z = h_0, \qquad P_{-i} Z = W h_1.$$

The fixed-sector interface costs zero; the interface allowing either of these two sectors costs $\mathrm{H}(W)$*, even in a single round.*

Proof. The first readout is constant; the second uniquely identifies W because $h_1 \neq 0$. Seed W generates both readouts, and the second gives the lower bound. □

Uniform W on m values gives a gap of $\log_2 m$. The hidden information resides in Z, not in the constant output h_0.

Why sequential access is essential. Joint reconstructability of a list of queries is not sufficient if only one query can ever be asked. Let U, V be independent fair bits and $Z = U h_0 + V h_1$. One round allowing either sector query costs only one bit: a single fair seed B can answer $B h_0$ or $B h_1$ as requested. The separate marginals are correct, though this seed does not reproduce their joint independence. Two rounds allowing both queries require two bits, because the fixed policy requesting both reconstructs (U, V). This demonstrates why the experimental interface, including its horizon, belongs in the theorem.

6 Safety and the viable subtree

In a finite controlled tree, let B be a set of forbidden histories closed under extension. Define $V(h)$ as the minimum probability of visiting B before termination in the continuation tree rooted at h. Backward induction gives

$$V(h) = \begin{cases} 1, & h \in B, \\ 0, & h \notin B \text{ is terminal}, \\ \min_a \sum_y p(y \mid h, a) V(hay), & \text{otherwise}. \end{cases}$$

Call h viable if $V(h) = 0$. At a viable nonterminal history retain

$$A_0(h) = \{a : \ p(y \mid h, a) > 0 \Rightarrow V(hay) = 0 \text{ for all } y\}.$$

This set is nonempty. If the root is viable, let T_0 be the tree reached using these actions and positive-probability edges. Let Π_0 contain all policies of the original tree with zero failure probability.

Theorem 6.1 (Exact safety restriction)**.** *Under these assumptions,*

$$\boxed{C_{\Pi_0}(T) = C(T_0).}$$

Proof. Every policy on T_0 extends arbitrarily outside T_0 to a safe policy on T. Restricting a simulator exact for all safe policies therefore gives a simulator on T_0. If its decoder assigns a deleted output at an unused seed–history pair, replace that output by any retained output: exactness guarantees the replacement changes no positive-probability run of any restricted policy. There are only finitely many policies and histories. Seed entropy is unchanged. Thus $C(T_0) \le C_{\Pi_0}(T)$.

Conversely, extend a decoder on T_0 arbitrarily to the remaining interfaces of T. Fix a safe policy π. Every history it reaches with positive probability has a zero-risk continuation. Hence at each such nonterminal history its action lies in $A_0(h)$. Define a policy on T_0 agreeing there with π, and choose arbitrary retained actions elsewhere. Induction on the horizon shows that its target law equals that of π. Run both decoders with the same seed. They agree until a first departure from those histories; exactness of the restricted run makes the probability of such a departure zero. The extension is therefore exact for π. Since this holds for every safe policy, taking infima gives the reverse inequality. □

This is an operational restriction, not a change to the full original laws. For example, a one-step tree with a deterministic safe action and a second action uniform on m forbidden outcomes has $C(T) = \log_2 m$ but $C_{\Pi_0}(T) = 0$. If the root is not viable, no safe policy exists. When the companion bound applies to T_0, it implies $L(T_0) \le C_{\Pi_0}(T) \le L(T_0) + c_\star$.

7 Invariants, quantum measurements, and the physical boundary

For a smooth controlled system $\dot{x} = F_a(x)$, a differentiable quantity U is conserved along every fixed-action trajectory when $dU_x(F_a(x)) = 0$ for every action and state. An intervention Θ_a preserves that quantity only when $U \circ \Theta_a = U$ is separately established. Commuting with the flow is insufficient: scaling the harmonic oscillator state by two commutes with its flow but multiplies $q^2 + p^2$ by four. Such conservation identities alone do not specify a probability law or a seed entropy.

For quantum experiments, a precise bridge can be specified at the level of instruments. Let ρ_h be the conditional density operator and let $\{\mathcal{I}_{a,y}\}_y$ be completely positive trace-nonincreasing maps whose sum is trace preserving. Then

$$p(y \mid h, a) = \operatorname{tr} \mathcal{I}_{a,y}(\rho_h), \qquad \rho_{hay} = \frac{\mathcal{I}_{a,y}(\rho_h)}{p(y \mid h, a)} \quad \text{when } p(y \mid h, a) > 0.$$

Finite action and outcome alphabets and a finite horizon produce a finite controlled tree. State updates incorporate measurement back-action; a passive writer does not automatically implement them. Probabilities on unreachable zero-probability histories require a specified convention if they are retained.

For a normalized two-path joint state $|\Psi\rangle = |\psi_1\rangle|d_1\rangle + |\psi_2\rangle|d_2\rangle$, with normalized detector states, tracing out the detector gives

$$p(x) = |\psi_1(x)|^2 + |\psi_2(x)|^2 + 2\,\mathrm{Re}\Big(\psi_1(x)\overline{\psi_2(x)}\langle d_2|d_1\rangle\Big).$$

This is standard quantum mechanics [6, 5], not a consequence of Shannon entropy. A finite detector partitions positions into recorded outcomes. Exact simulation refers to those outcome laws, not automatically to a complete continuous quantum field.

The defensible direction is therefore

$$\text{physical model and experimental protocol} \longrightarrow p(y \mid h, a) \longrightarrow T \longrightarrow C(T).$$

A displayed Lagrangian alone omits the preparation and measurement specification. No Lagrangian, Born rule, gravitational dynamics, or consciousness mechanism is derived here. Classical centralized simulation of a temporal interface does not establish spatially local hidden variables or efficient computation.

8 A conditional estimate is not a proof of RH

The supplied Fourier manuscript also proposes a conditional bound of the form

$$\frac{Q(\tau, L) - E_0 L}{L} \geq A\tau^4 - B\tau^3 - D, \qquad A > 0, \quad B, D \geq 0,$$

valid uniformly for $L \geq 1$ and $\tau \geq \tau_0$. Define

$$\tau_\star = \max\left\{\tau_0, 1, \frac{2B}{A}, \left(\frac{4D}{A}\right)^{1/4}\right\}.$$

Then for $\tau \geq \tau_\star$ the right side is strictly positive: it is at least $A\tau^4/2 - D$, which is at least $D > 0$ if $D > 0$, and is positive if $D = 0$. This is a valid elementary implication. It does not prove the assumed bridge or quartic coercivity estimates. Fourier-sector decomposition and seed entropy do not supply those estimates. No proof of the Riemann hypothesis is claimed in this synthesis.

9 Conclusions and the remaining mathematical question

The precise content of "unfolding" is a sequence of choices and responses whose complete history is generated by one common seed and decoder. Reconstruction turns an entropy lower bound into an exact optimum. Fourier sectors give a concrete family of observation interfaces, and safety gives a concrete restriction of the policies to be supported.

The main operational identities established here are

$$\boxed{\begin{aligned} C_{\text{reconstructing interface}} &= \mathrm{H}(Z), \\ C_{\text{four sequential sectors}} - C_{\text{fixed sector}} &= \mathrm{H}(Z \mid P_1 Z), \\ C_{\text{safe policies}}(T) &= C(T_0). \end{aligned}}$$

Each equality has its stated finite-model assumptions. None equates information with matter, probability with physical energy, or entropy with computational time. In particular, no implication $P = NP$ follows. Love, intention, and metaphysical potential can motivate a question but are not variables identified by these proofs.

A concrete further problem is the cost of supporting all policies with failure probability at most $\varepsilon > 0$. This class can couple choices across branches through a global risk budget, so a fixed local pruning rule need not describe it. A sharp structural characterization and corresponding entropy bounds remain research targets, not results of this paper.

Proof and attribution status. The finite proofs above are mathematical arguments, not a new Lean certificate. The Hermite and entropy ingredients are classical. The synthesis and examples are not accompanied by a comprehensive priority search, and no claim that they are previously unknown is made. A previous inspection of repository commit `f54c6e6` found the writer formalization weaker than the full adaptive claim; this manuscript does not audit subsequent revisions. The companion bound (1) remains explicitly separated from the direct proofs.

References

[1] J. Mikulík, *Finite Shannon Theorem*, companion repository, `https://github.com/jendamikulik/finite-Shannon-theorem`.

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[4] M. L. Puterman, *Markov Decision Processes*, Wiley, 1994.

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[6] R. P. Feynman, R. B. Leighton and M. Sands, *The Feynman Lectures on Physics*, Vol. III, Ch. 3, `https://www.feynmanlectures.caltech.edu/III_03.html`.

[7] S. L. Onweller, *ONE*, supplied edition `ONE_E24_Gabriels_Horn_Integrated_Pass7.pdf`, writer updates in Theorem PS.1, printed p. 17. Cited only for the specified update construction.